

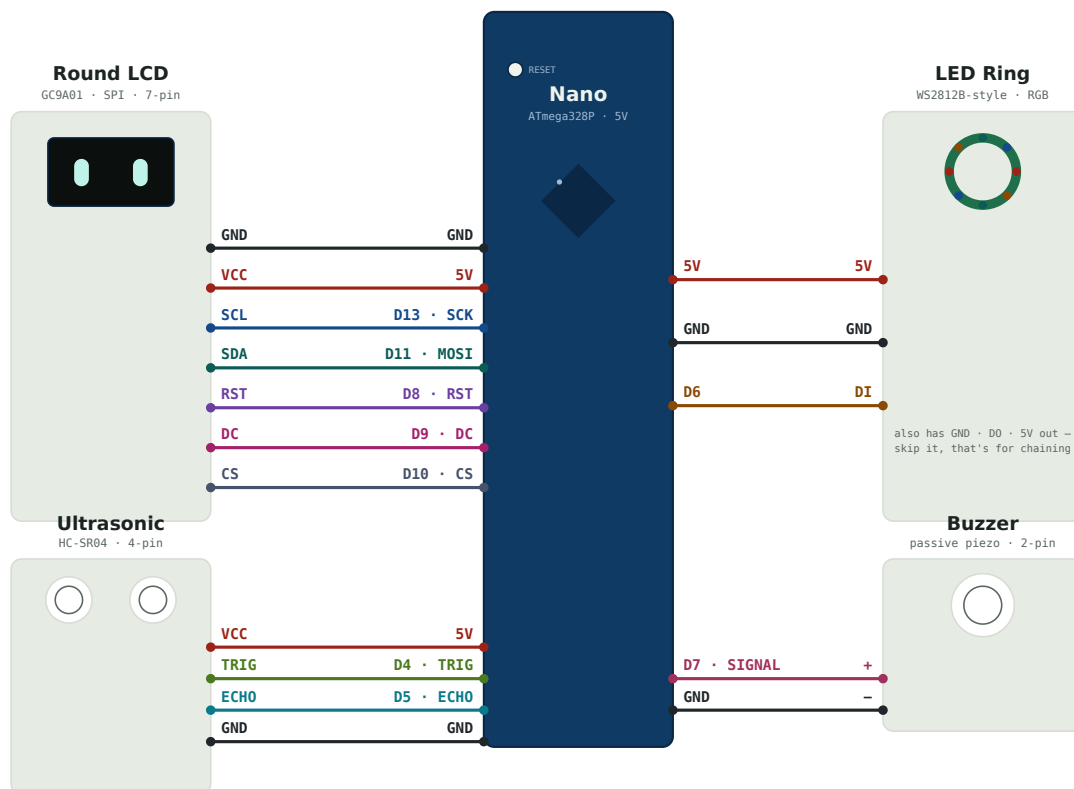
Nano Wiring

Round GC9A01 display, WS2812B-style ring, HC-SR04 ultrasonic sensor, and a piezo buzzer, all on one classic Arduino Nano (ATmega328P). Confirmed from the board itself: Mini-USB, single ATmega328P chip, no wireless module.

2KB RAM — can't hold a full 240×240 framebuffer (115KB). Draw window-by-window with `Adafruit_GC9A01A` + `Adafruit_GFX`, same direct-draw approach used on the Pico, not a buffered one.

5V logic confirmed fine for the display per its packaging, and native for the sensor and buzzer too — no level shifter needed anywhere on this board.

Full rewrite — Arduino uses C++ (Arduino IDE/CLI), not MicroPython. None of the Pico scripts carry over as-is.



D13/D11 are the Nano's hardware SPI (SCK/MOSI); D8–D10 are plain GPIO for RST/DC/CS. D6 drives the ring. D4/D5 fire and read the sensor's ping. D7 drives the buzzer. Every GND shown is the same net.

■ GND ■ Power – 5V ■ SCK – SPI clock ■ MOSI – SPI data ■ RST – reset
■ DC – data/command ■ CS – chip select ■ DI – LED data-in
■ TRIG – sensor ping out ■ ECHO – sensor pulse back ■ SIGNAL – buzzer tone

Before you power it on

- Every GND pin is the same net — every part here can land on whichever GND is closest, they don't need to share one physical pin.
- **Ground first, always double-check it.** A loose GND reads as "everything's broken" with no error anywhere — the single most time-consuming fault to chase on this whole build.
- Wire the ring's **DI** side only (GND · DI · 5V). Leave GND · DO · 5V alone.
- HC-SR04's TRIG and ECHO aren't interchangeable — TRIG is an output from the Nano, ECHO is an input to it. Swapped, you'll get silence or garbage readings, no error.
- Passive piezo buzzer has no polarity — either leg to D7, the other to GND, it doesn't matter which way round.
- Nano's onboard 3.3V regulator is weak (~50mA) — everything here runs off **5V**, not 3V3.
- D13 also drives the Nano's onboard **L** LED — harmless here, but expect it to flicker along with SCK traffic.
- Unplug USB before changing wiring, check for shorts before plugging back in.